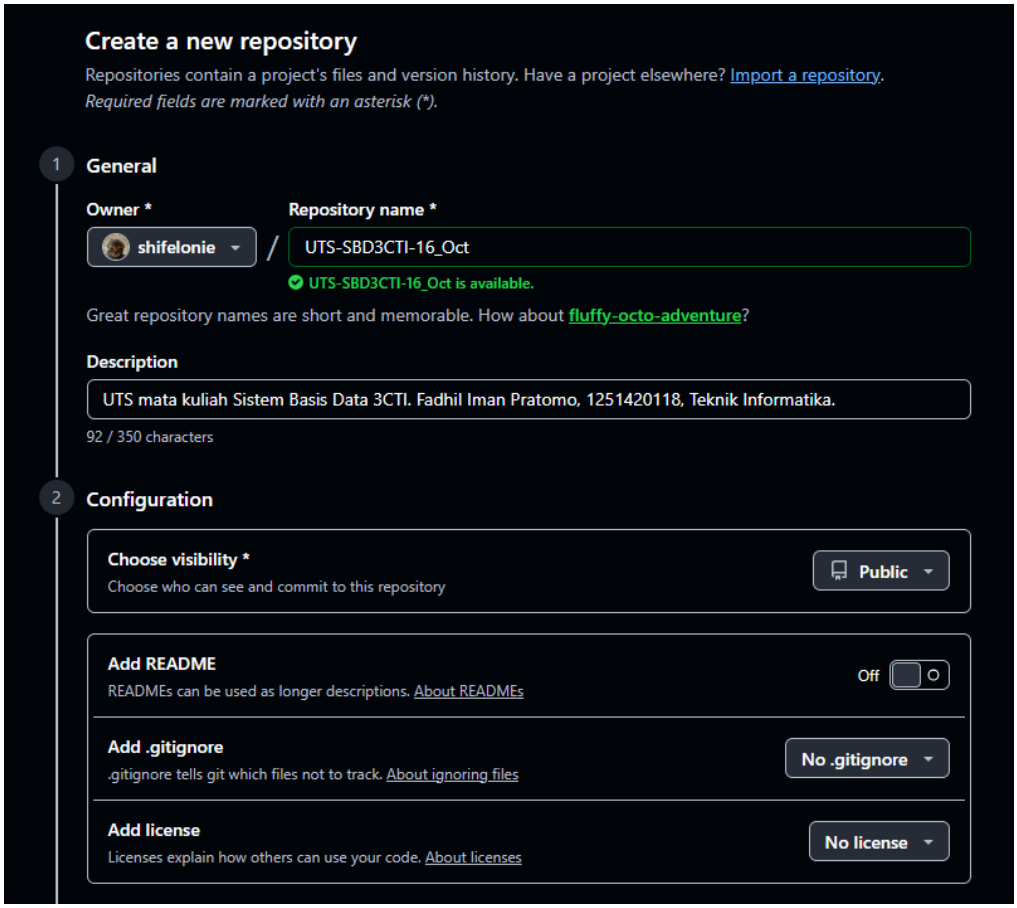
FADHIL IMAN PRATOMO | 1251420118 | TEKNIK INFORMATIKA

UTS SISTEM BASIS DATA 3CTI

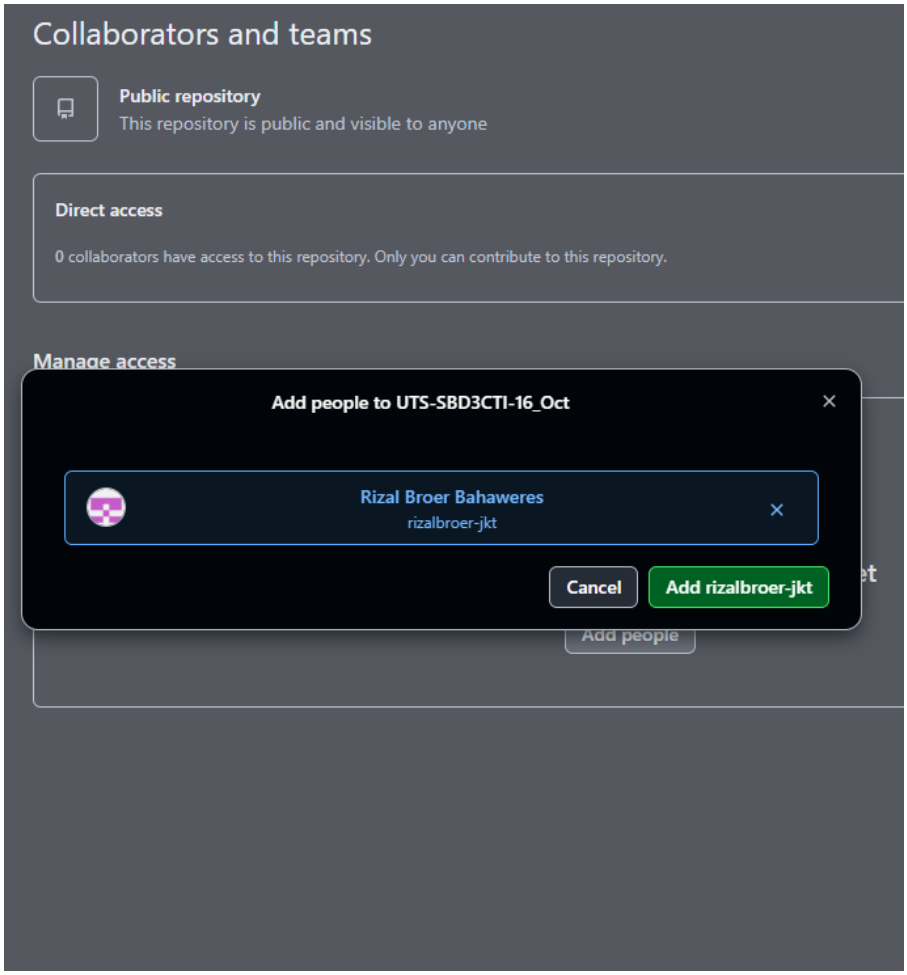
1.

Membuat repository pada akun Github.



2.

Menambahkan akun Github Pak Broer sebagai *Collaborators*.



3.

Membuat database **dreamhome** terlebih dahulu

```
postgres=# CREATE DATABASE dreamhome;
CREATE DATABASE
```

6.10

4.

Membuat table **Hotel** dengan menggunakan fitur integrity enhancement pada PostgreSQL. Pada kasus kali ini, user akan menambahkan tahapan dengan membuat sebuah extension yang bernamakan **btree_gist** untuk membantu menjawab soal **6.11** poin (e) dan (f) pada soal **6.11** nanti.

```
dreamhome=# CREATE EXTENSION IF NOT EXISTS btree_gist;
CREATE EXTENSION
dreamhome=# CREATE TABLE Hotel (hotelNo VARCHAR(10) PRIMARY KEY, hotelName VARCHAR(50) NOT NULL, city VARCHAR(50) NOT NULL);
CREATE TABLE
dreamhome=#
```

6.11

5.

Sekarang, user diminta untuk membuat table **Room**, **Booking** & **Guest** dengan bantuan fitur integrity enhancement juga, dengan masing-masing table memiliki ketentuan-ketentuan nya masing-masing.

Table **Guest**:

```
dreamhome=# CREATE TABLE Guest (
dreamhome(#     guestNo VARCHAR(10) PRIMARY KEY,
dreamhome(#     guestName VARCHAR(50) NOT NULL,
dreamhome(#     guestAddress VARCHAR(100) NOT NULL
dreamhome(# );
CREATE TABLE
dreamhome=#
```

Table **Room** dengan ketentuan:

(a) atribut type harus antara Single, Double, atau Family
(b) atribut price harus antara £10 hingga £100
(c) atribut roomNo harus diantara 1 hingga 100

```
dreamhome=# CREATE TABLE Room (
dreamhome(#     roomNo INT NOT NULL CHECK (roomNo BETWEEN 1 AND 100),
dreamhome(#     hotelNo VARCHAR(10) NOT NULL,
dreamhome(#     type VARCHAR(10) NOT NULL CHECK (type IN ('Single', 'Double', 'Family')),
dreamhome(#     price DECIMAL(5,2) NOT NULL CHECK (price BETWEEN 10 AND 100),
dreamhome(#     PRIMARY KEY (roomNo, hotelNo),
dreamhome(#     FOREIGN KEY (hotelNo) REFERENCES Hotel(hotelNo) ON UPDATE CASCADE ON DELETE CASCADE
dreamhome(# );
CREATE TABLE
dreamhome=#
```

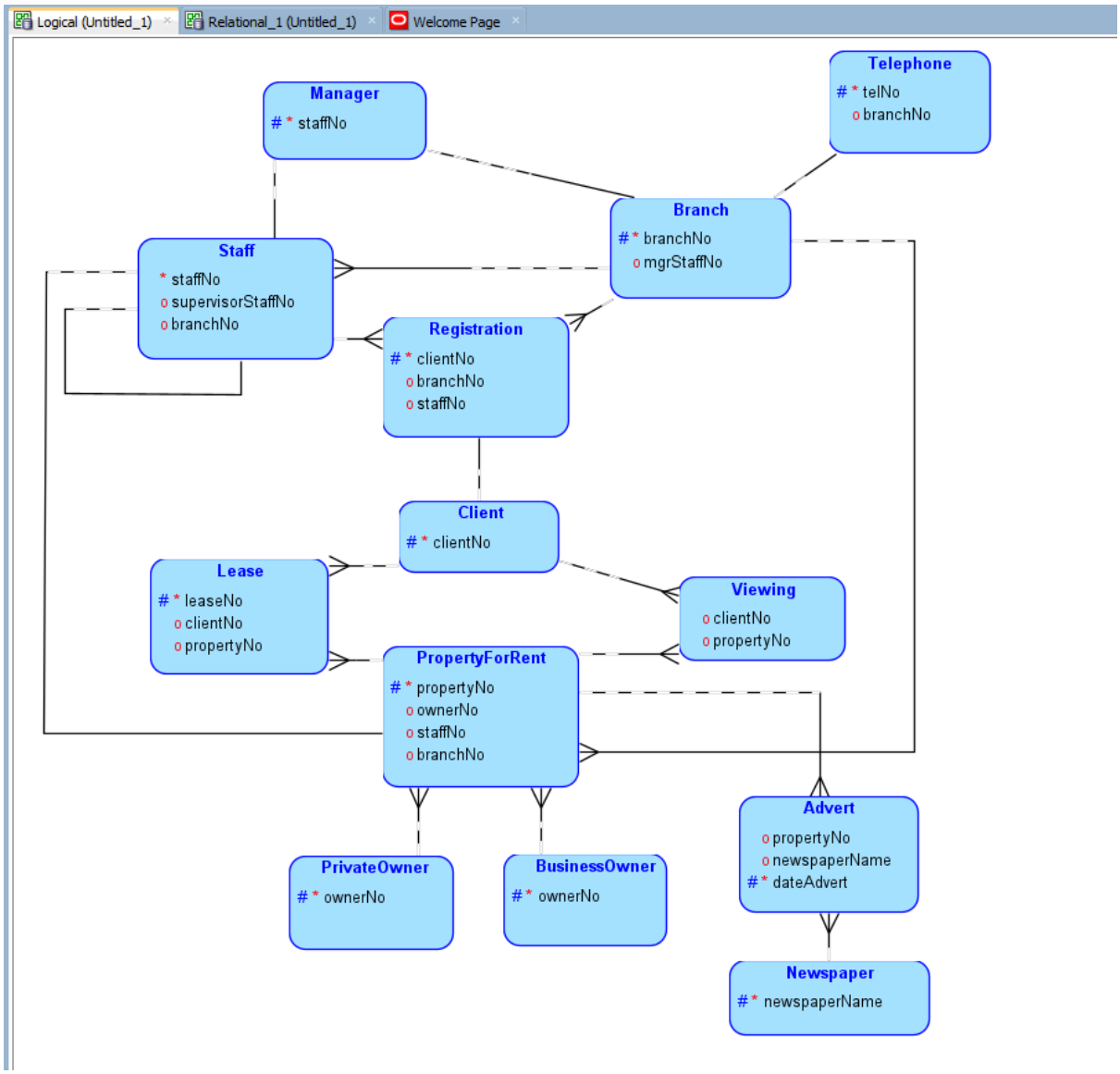
Table **Booking** dengan ketentuan:

(d) atribut dateFrom dan dateTo harus memiliki tanggal setelah dari tanggal hari ini (greater than today's date)
(e) ketentuan room yang sama tidak bisa di book sebanyak 2 kali dalam waktu yang bersamaan (CONSTRAINT no_double_booking)
(f) ketentuan guest yang sama tidak bisa memiliki booking yang saling menumpuk satu sama lain (CONSTRAINT no_overlapping_guest)

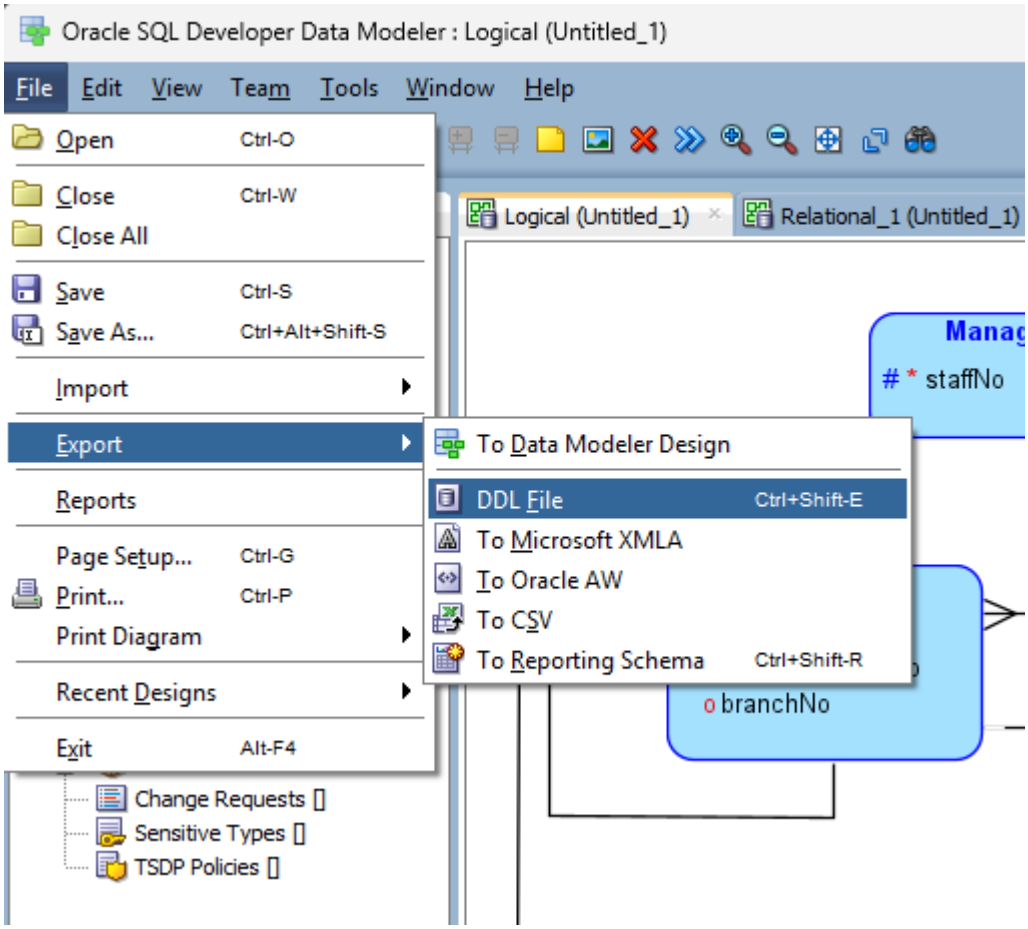
```
dreamhome=# CREATE TABLE Booking (
dreamhome(#     hotelNo VARCHAR(10) NOT NULL,
dreamhome(#     guestNo VARCHAR(10) NOT NULL,
dreamhome(#     dateFrom DATE NOT NULL CHECK (dateFrom > CURRENT_DATE),
dreamhome(#     dateTo DATE NOT NULL CHECK (dateTo > CURRENT_DATE AND dateTo > dateFrom),
dreamhome(#     roomNo INT NOT NULL,
dreamhome(#     PRIMARY KEY (hotelNo, guestNo, dateFrom),
dreamhome(#     FOREIGN KEY (hotelNo) REFERENCES Hotel(hotelNo) ON UPDATE CASCADE ON DELETE CASCADE,
dreamhome(#     FOREIGN KEY (guestNo) REFERENCES Guest(guestNo) ON UPDATE CASCADE ON DELETE CASCADE,
dreamhome(#     FOREIGN KEY (roomNo, hotelNo) REFERENCES Room(roomNo, hotelNo) ON UPDATE CASCADE ON DELETE CASCADE,
dreamhome(#
dreamhome(#     CONSTRAINT no_double_booking EXCLUDE USING gist (
dreamhome(#         roomNo WITH =,
dreamhome(#         hotelNo WITH =,
dreamhome(#         daterange(dateFrom, dateTo) WITH &&
dreamhome(#     ),
dreamhome(#
dreamhome(#     CONSTRAINT no_overlapping_guest EXCLUDE USING gist (
dreamhome(#         guestNo WITH =,
dreamhome(#         daterange(dateFrom, dateTo) WITH &&
dreamhome(#     )
dreamhome(# );
CREATE TABLE
dreamhome=#
```

Data Modeler

Logical Model:



Jika sudah, kita akan men-generate file DDL dari Logical Model diatas menjadi query yang dapat kita execute pada PostgreSQL.



Physical Model (Dump DDL hasil generate Logical pada PostgreSQL):

DDL yang akan di-dump:

```
CREATE TABLE Branch (
    branchNo VARCHAR(5) PRIMARY KEY,
    mgrStaffNo VARCHAR(5) -- ini bakalan di-alter nanti karena hubungan circular constraint
);

CREATE TABLE Telephone (
    telNo VARCHAR(15) PRIMARY KEY,
    branchNo VARCHAR(5) NOT NULL,
    FOREIGN KEY (branchNo) REFERENCES Branch(branchNo)
);

CREATE TABLE Staff (
```

```
    staffNo VARCHAR(5) PRIMARY KEY,
    supervisorStaffNo VARCHAR(5),
    branchNo VARCHAR(5) NOT NULL,
    FOREIGN KEY (supervisorStaffNo) REFERENCES Staff(staffNo),
    FOREIGN KEY (branchNo) REFERENCES Branch(branchNo)
);

-- foreign key untuk manajer cabang
ALTER TABLE Branch ADD CONSTRAINT fk_branch_mgr FOREIGN KEY (mgrStaffNo)
REFERENCES Staff(staffNo);

CREATE TABLE Manager (
    staffNo VARCHAR(5) PRIMARY KEY,
    FOREIGN KEY (staffNo) REFERENCES Staff(staffNo)
);

CREATE TABLE Client (
    clientNo VARCHAR(5) PRIMARY KEY
);

CREATE TABLE Registration (
    clientNo VARCHAR(5) PRIMARY KEY,
    branchNo VARCHAR(5) NOT NULL,
    staffNo VARCHAR(5) NOT NULL,
    FOREIGN KEY (clientNo) REFERENCES Client(clientNo),
    FOREIGN KEY (branchNo) REFERENCES Branch(branchNo),
    FOREIGN KEY (staffNo) REFERENCES Staff(staffNo)
);

CREATE TABLE PrivateOwner (
    ownerNo VARCHAR(5) PRIMARY KEY
```

```
);

CREATE TABLE BusinessOwner (
    ownerNo VARCHAR(5) PRIMARY KEY
);

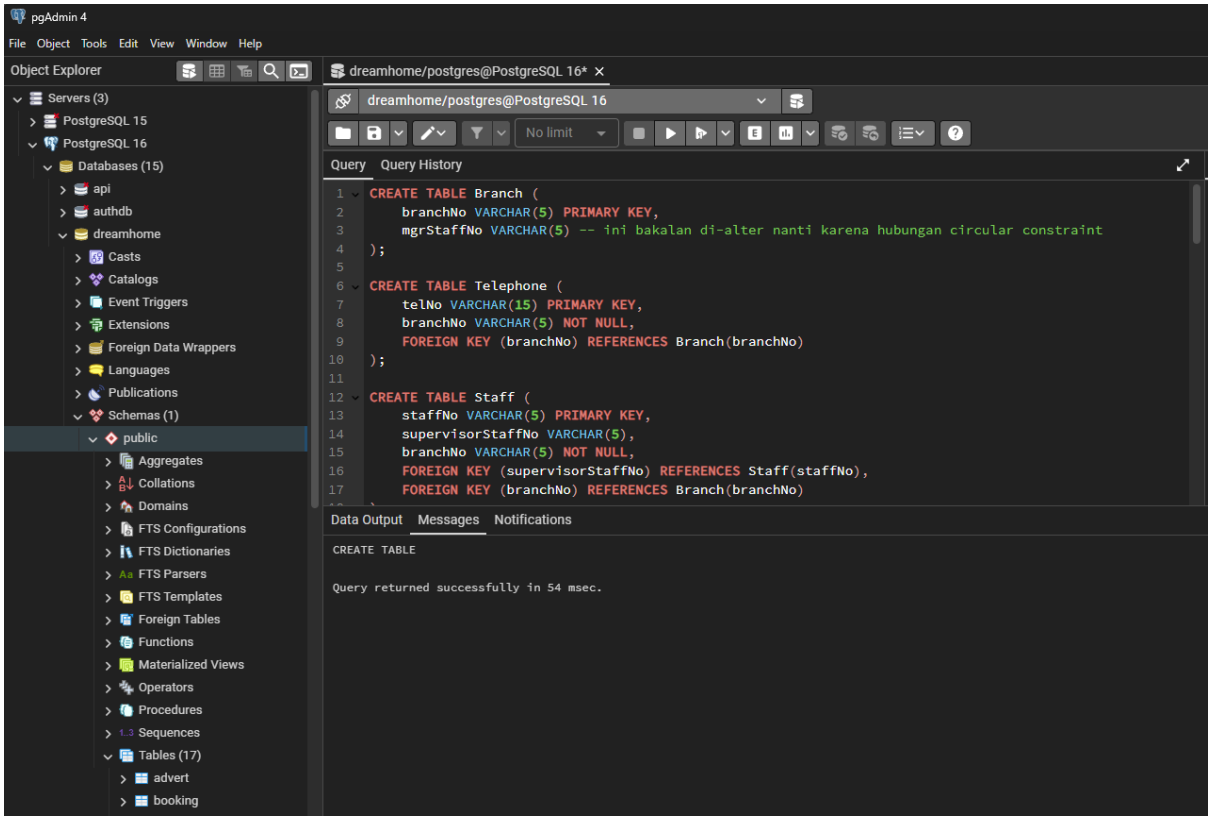
CREATE TABLE PropertyForRent (
    propertyNo VARCHAR(5) PRIMARY KEY,
    ownerNo VARCHAR(5) NOT NULL,
    staffNo VARCHAR(5),
    branchNo VARCHAR(5) NOT NULL,
    FOREIGN KEY (staffNo) REFERENCES Staff(staffNo),
    FOREIGN KEY (branchNo) REFERENCES Branch(branchNo)
);

CREATE TABLE Viewing (
    clientNo VARCHAR(5),
    propertyNo VARCHAR(5),
    PRIMARY KEY (clientNo, propertyNo),
    FOREIGN KEY (clientNo) REFERENCES Client(clientNo),
    FOREIGN KEY (propertyNo) REFERENCES PropertyForRent(propertyNo)
);

CREATE TABLE Lease (
    leaseNo VARCHAR(5) PRIMARY KEY,
    clientNo VARCHAR(5) NOT NULL,
    propertyNo VARCHAR(5) NOT NULL,
    FOREIGN KEY (clientNo) REFERENCES Client(clientNo),
    FOREIGN KEY (propertyNo) REFERENCES PropertyForRent(propertyNo)
);
```

```
CREATE TABLE Newspaper (
    newspaperName VARCHAR(50) PRIMARY KEY
);

CREATE TABLE Advert (
    propertyNo VARCHAR(5),
    newspaperName VARCHAR(50),
    dateAdvert DATE,
    PRIMARY KEY (propertyNo, newspaperName, dateAdvert),
    FOREIGN KEY (propertyNo) REFERENCES PropertyForRent(propertyNo),
    FOREIGN KEY (newspaperName) REFERENCES Newspaper(newspaperName)
);
```



Berikut ini adalah table-table yang telah berhasil dibuat pada database **dreamhome**:

```
dreamhome=# \dt
              List of relations
 Schema |       Name        | Type  |  Owner
--------+-------------------+-------+----------
 public | advert            | table | postgres
 public | booking           | table | postgres
 public | branch            | table | postgres
 public | businessowner     | table | postgres
 public | client            | table | postgres
 public | guest             | table | postgres
 public | hotel             | table | postgres
 public | lease             | table | postgres
 public | manager           | table | postgres
 public | newspaper         | table | postgres
 public | privateowner      | table | postgres
 public | propertyforrent   | table | postgres
 public | registration      | table | postgres
 public | room              | table | postgres
 public | staff             | table | postgres
 public | telephone         | table | postgres
 public | viewing           | table | postgres
(17 rows)
```